

Searching Binary Embeddings

Bitplanes and Prefix Backward Walk

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Abstract

I study three ways to search binary document vectors after selecting clusters: a full scan, Bitplanes, and Backward Walk. I compare their work, memory use and retrieval time using the same documents and queries. The experiments examine when the extra index work saves enough document scoring operations to reduce latency at a required recall.

1 Introduction

Throughout this report, a document scoring operation is the computation; its score value is the resulting number, such as 1.9.

1.1 Exa method and end to end pipeline

Let's first understand the "original" pipeline that I am trying to improve upon.

Note that the architecture I am referring to here is based on the pipeline described in Exa's article [1]. I am aware that Exa's actual production system may have changed or improved since then, but for the purpose of this report, let's treat the architecture described in that article as the original baseline that we compare against.

Detail: In this pipeline, document embeddings are first shortened using Matryoshka representations, converted into binary signs, and assigned to clusters. At query time, the query is encoded, a subset of nearby clusters is selected, and the documents inside those clusters undergo scoring operations using the floating-point query. The top candidates can then be reranked using the original uncompressed vectors.

The main part that I am trying to improve in this report is the search stage after cluster selection. The two new approaches introduced later, Bitplanes and Backward Walk, replace this step while keeping the rest of the pipeline largely the same.

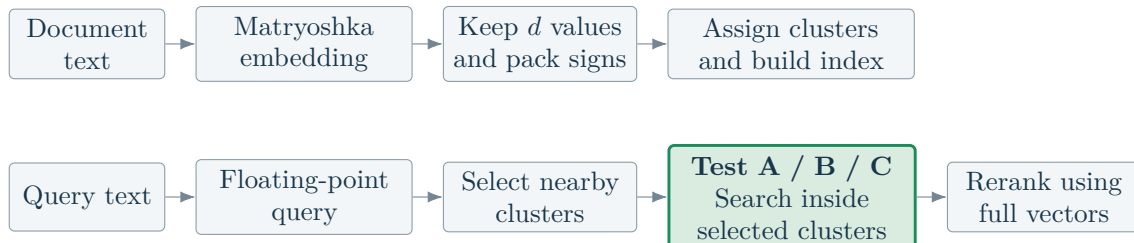


Figure 1: Green marks the part we aim to improve: document search inside the selected clusters, using A, B or C. Reranking is shown as a separate step.

1.2 Idea background

When I took a closer look at the pipeline, one part that particularly stood out to me was the comparison between the floating-point query and the binary-quantized document.

Consider a simple example:

$$q = [0.7, 0.5, -0.4, -0.6, 0.3]$$

and suppose a document is represented using binary signs, where 1 means +1 and 0 means -1. For example:

$$\text{Document} = 11000$$

The score value is calculated by taking the dot product between the query values and the document signs. So for 11000, the score value is

$$0.7(+1) + 0.5(+1) + (-0.4)(-1) + (-0.6)(-1) + 0.3(-1) = 1.9.$$

Now, something interesting follows from this. Just by looking at the signs of the query, we already know what the ideal binary document would look like.

For every positive query value, we would prefer the corresponding document bit to be 1, and for every negative query value, we would prefer it to be 0. Therefore, for the query above, the ideal binary pattern is

$$11001.$$

This document would achieve the maximum possible score value:

$$0.7 + 0.5 + 0.4 + 0.6 + 0.3 = 2.5.$$

From here, we can think about every other document in terms of how far it is from this ideal pattern. More importantly, not every mismatch has the same cost. If we mismatch the first coordinate, where $|q_1| = 0.7$, the score value decreases by 1.4. But if we mismatch the last coordinate, where $|q_5| = 0.3$, the score value only decreases by 0.6.

In more formal terms, let

$$Q = \sum_i |q_i|$$

be the score value of the ideal binary pattern. If a document b disagrees with the query sign at some coordinates, define its mismatch penalty as

$$p = \sum_{i: b_i \neq \text{sign}(q_i)} |q_i|.$$

Then its score value can be written as

$$S(q, b) = Q - 2p. \tag{1}$$

This observation is the main idea behind the approaches explored in this report. Instead of treating every binary document equally and performing a scoring operation for each one, can we use the query itself to guide the search toward documents that are closer to its ideal binary pattern, while avoiding unnecessary document scoring operations?

1.3 Exploring new methods

From the previous section, we know two useful things before we even start searching:

We know what the ideal binary document looks like from the signs of the query. We know that some bits matter more than others, based on the magnitude $|q_i|$.

The next question is therefore:

Can we build an index that uses this information directly, instead of treating every document inside the selected clusters equally?

This led me to explore two approaches.

The first, Bitplanes, starts from a large set of candidate documents and progressively narrows it down using the query.

The second, Backward Walk, approaches the problem from the opposite direction: it starts from a very specific binary pattern and progressively relaxes the constraints until enough useful candidates are found.

1.3.1 Bitplanes

Now that we know what the ideal document looks like based on the query, one idea is to build a data structure that makes filtering by individual bits very cheap.

Normally, binary documents are stored row by row:

ID 2: 10100
 ID 3: 11001
 ID 4: 11000

Instead, we can additionally store an inverted representation of the same bits:

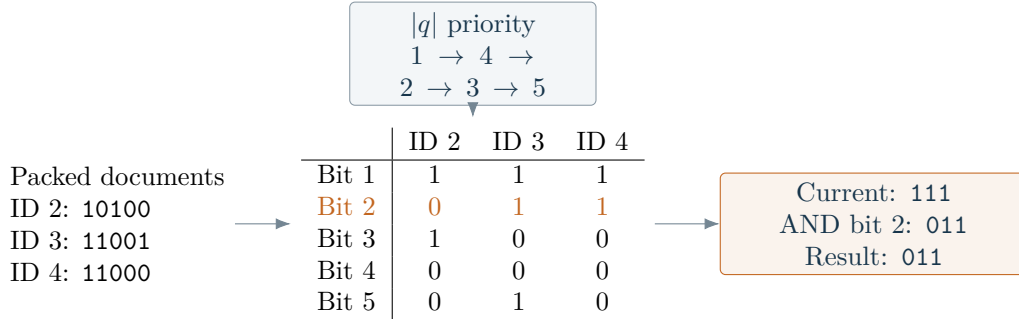


Figure 2: The same document signs, stored in an extra layout for bitwise filtering.

Each row above is what I call a bitplane: one bit position across many documents.

Suppose the query is

$$q = [0.7, 0.5, -0.4, -0.6, 0.3].$$

From the magnitude of the query values, we can prioritize the bits approximately as

$$1, 4, 2, 3, 5,$$

because

$$0.7 > 0.6 > 0.5 > 0.4 > 0.3.$$

Now suppose our current candidate mask is

$$111$$

meaning that IDs 2, 3 and 4 are all still possible candidates.

For bit 2, the query prefers 1. The corresponding bitplane is

$$011$$

so keeping the documents that match the query is simply

$$111 \& 011 = 011.$$

We immediately reduce the candidate set to IDs 3 and 4.

This is attractive because the filtering itself consists mostly of bitwise operations, which modern CPUs can perform very efficiently.

A simplified first version of the idea might therefore look like:

```

def full_score(query, code):
    return sum(q if bit else -q for q, bit in zip(query, code))

def greedy_search(query, cluster, k, leaf_size):
    order = sorted(range(len(query)), key=lambda i: -abs(query[i]))
    active = cluster.full_mask
    for bit in order:
        plane = cluster.planes[bit]
        preferred = plane if query[bit] >= 0 else ~plane
        active &= preferred
        if active.bit_count() <= leaf_size:
            break

    best = TopK(k)
    for row_id, code in cluster.rows_in_mask(active):
        best.add(row_id, full_score(query, code))
    return best.results()

```

However, this simple version has a major problem.

PROBLEM 1: Greedily following the preferred bits can miss the best document

Consider:

$$q = [0.6, 0.5, 0.4, 0.3].$$

The ideal binary document is 1111.

Suppose the database only contains:

Document A: 0111

Document B: 1000

If we greedily start with the first bit, the query prefers 1. We would therefore keep Document B and throw away Document A.

But their complete score values are:

$$S(A) = -0.6 + 0.5 + 0.4 + 0.3 = 0.6$$

while

$$S(B) = 0.6 - 0.5 - 0.4 - 0.3 = -0.6.$$

So Document A is actually better.

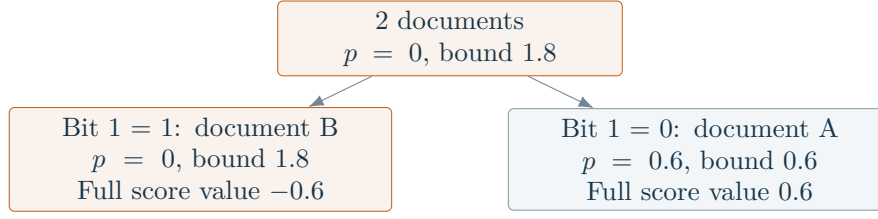
The problem is that matching one important bit does not guarantee that the remaining bits will also match.

Solution: branching

Instead of discarding the alternative, we keep both branches.

The branch matching the preferred bit receives no additional mismatch penalty. The alternative branch is still kept, but its penalty increases by $|q_i|$.

We then explore the branches in order of smallest accumulated penalty. This gives us a recursive search tree rather than one greedy path.



After the scoring operation for B, the other bound is $0.6 > -0.6$.
We must explore A; the first choice was not enough.

Figure 3: Keep the alternative until its bound on score values proves it cannot improve the result.

Stopping condition

Recall:

$$S(q, b) = Q - 2p, \quad Q = \sum_i |q_i|.$$

If a branch has already accumulated mismatch penalty p , any document below it must have penalty at least p .

Therefore,

$$U_{\text{branch}} = Q - 2p$$

is an upper bound on the score value of every document inside that branch.

For example, suppose $Q = 2.5$ and we are looking for the top two results. We have already found 2.5 and 1.9, where 1.9 is the current top-2 threshold.

Now consider an unexplored branch with $p = 0.5$. Its absolute best possible score value is

$$2.5 - 2(0.5) = 1.5.$$

Since $1.5 < 1.9$, nothing inside that branch can enter our current top two. We can safely discard the entire subtree.

So:

$$Q - 2p < \text{current } K\text{th score value} \Rightarrow \text{prune branch}$$

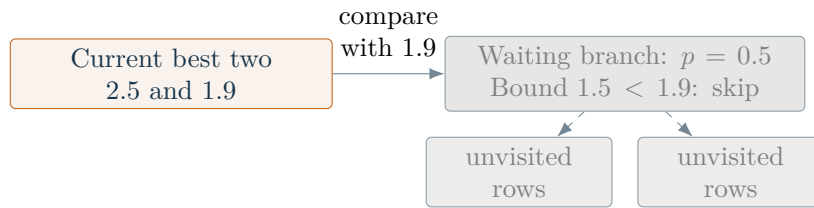


Figure 4: The whole gray subtree can be skipped because even its best possible score value is too low.

If the best remaining branch in the queue already fails this test, then every other branch is worse and the entire search can stop.

The resulting algorithm is approximately:

```

from heapq import heappop, heappush
from itertools import count

def bitplane_search(query, clusters, k, leaf_size):
    order = sorted(range(len(query)), key=lambda i: -abs(query[i]))
    ideal_score = sum(abs(q) for q in query)
    sequence = count()
    queue = []
    for cluster in clusters:
        if cluster.full_mask:
            heappush(queue, (0, next(sequence), 0,
                             cluster.full_mask, cluster))

    best = TopK(k)
    while queue:
        penalty, _, depth, active, cluster = heappop(queue)
        upper_bound = ideal_score - 2 * penalty
        if best.full() and upper_bound < best.worst_score():
            break
        if active.bit_count() <= leaf_size or depth == len(query):
            for row_id, code in cluster.rows_in_mask(active):
                best.add(row_id, full_score(query, code))
            continue

        bit = order[depth]
        plane = cluster.planes[bit]
        preferred = plane if query[bit] >= 0 else ~plane
        matching = active & preferred
        # Keep the other half too; its remaining bits may be better.
        children = ((matching, 0), (active ^ matching, abs(query[bit])))
        for mask, extra_penalty in children:
            if mask:
                heappush(queue, (penalty + extra_penalty,
                                 next(sequence), depth + 1, mask, cluster))

    return best.results()

```

This is essentially the branch-and-bound Bitplane search used in the current experiments.

PROBLEM 2: Bitwise operations are cheap, but the bitmap itself can still be large

There is another issue.

A bitwise AND sounds almost free, but it is not literally one operation when the cluster contains thousands or millions of documents.

Suppose one cluster contains 6400 documents. With 64-bit machine words, one bitplane contains

$$6400/64 = 100$$

machine words.

So one split roughly requires us to loop over those 100 words.

Even worse, suppose after several splits only 80 documents remain. The physical mask can still be 100 words wide. So the next split may still need to process those same 100 words.

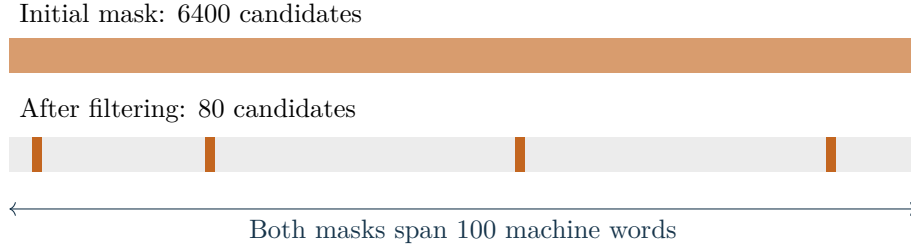


Figure 5: Fewer active document positions do not make this dense mask physically shorter.

Bitmap traversal cost. In general, for a cluster containing n documents, one dense bitplane operation costs roughly $O(\lceil n/64 \rceil)$ word operations. If we visit J split nodes, this becomes approximately $O(J \lceil n/64 \rceil)$, before even counting queue operations and the final document scoring operations. This was one of the important findings from the earlier experiments: reducing the number of documents that undergo a scoring operation does not automatically mean reducing total query time, because the bitmap traversal itself can become expensive.

Solution 1: Parameter tuning. This problem is technically something we can optimize. For example, we can tune cluster size during coarse routing; number of clusters probed; how deep we split; leaf size; when we switch back to ordinary document scoring operations. So Bitplanes is not fundamentally broken. The question is finding the point where scoring operations avoided are worth more than additional bitmap traversal.

Solution 2: Other approach. However, thinking about this cost led me to a second idea. Bitplanes starts with a large set of documents and repeatedly narrows it down. What if we instead did the opposite? If Bitplanes can be thought of as bottom-up, then Backward Walk can be thought of as top-down: start from the most specific binary pattern we can describe and progressively broaden the search only when necessary.

1.3.2 Backward Walk

Backward Walk builds on the same observation:

Given a query, we already know what its ideal binary document should look like.

However, instead of starting with thousands of documents and filtering them using bitplanes, we try to start directly from a small bucket of documents that already looks similar to the query.

Suppose the binary query begins with 101101001011....

Using all 256 bits as a lookup key would be far too specific. With a large binary space, the probability that an exact 256-bit pattern actually exists in the database is extremely small.

Instead, suppose we choose a prefix length x . For example, the first 16 bits:

1011010010110101

During index construction, documents sharing this prefix are placed in the same bucket or contiguous range.

At query time, we directly look up this prefix. If it contains enough useful documents, we perform their scoring operations using the complete 256-bit scoring function. If it is empty, or contains too few candidates, we remove one constraint:

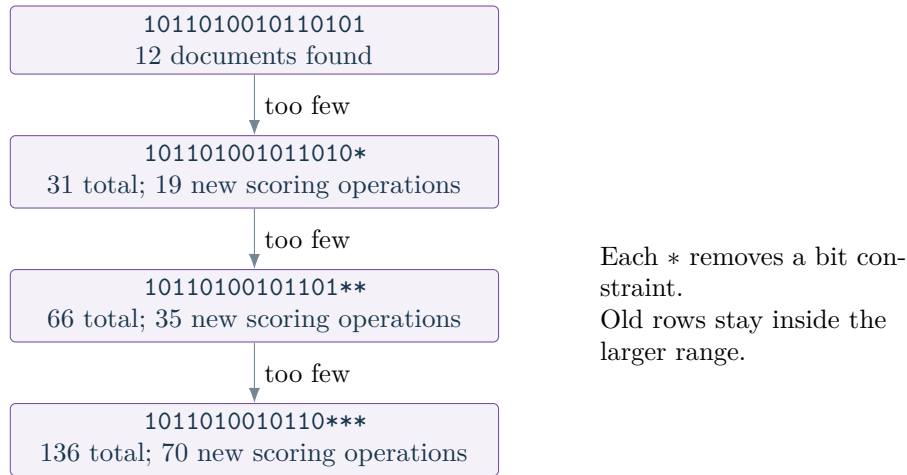


Figure 6: Illustrative counts: $12 + 19 + 35 + 70 = 136$ distinct document scoring operations, not $12 + 31 + 66 + 136$.

Every step moves to a larger candidate set.

This is why I call the method Backward Walk: instead of progressively adding constraints to a large candidate set, we begin from a very specific location and progressively walk backward toward more general prefixes.

Note that this bit prefix already has an exponential property: if the bits are independent and equally likely to be 0 or 1, each removed bit doubles the expected number of matching documents.

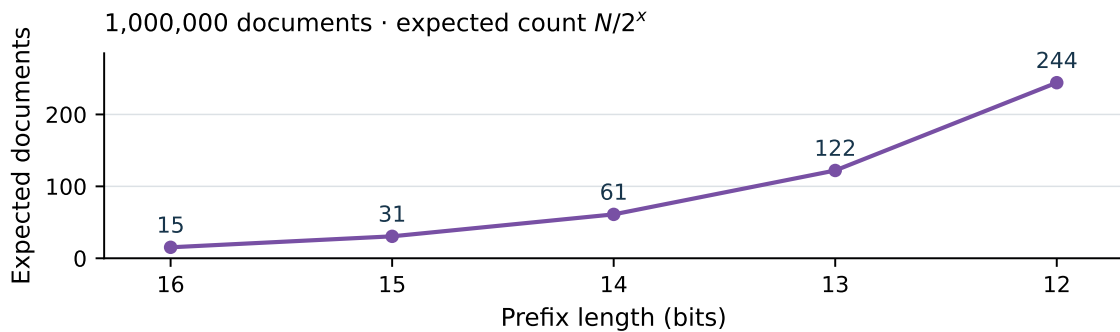


Figure 7: Expected counts for one global cluster; labels are rounded. This estimates rows, not recall.

With a trie or documents ordered by their prefix, the previous range is contained inside the parent range. We only need to process the newly introduced documents.

A simplified query algorithm becomes:

```
def backward_walk(query, clusters, k, start_depth, candidate_target):
    signs = tuple(q >= 0 for q in query)
    previous = [None] * len(clusters)
    best = TopK(k)
    scored = 0
    for depth in range(start_depth, -1, -1):
        for index, cluster in enumerate(clusters):
            left, right = cluster.prefix_range(signs[:depth])
            if previous[index] is None:
                new_ranges = [(left, right)]
            else:
                old_left, old_right = previous[index]
                # Only the new rows beside the old range need scoring operations.
                new_ranges = [(left, old_left), (old_right, right)]
            for start, stop in new_ranges:
                for row_id, code in cluster.rows_in_range(start, stop):
                    best.add(row_id, full_score(query, code))
                scored += stop - start
            previous[index] = (left, right)

        # Finish this depth in every cluster before testing the target.
        if candidate_target > 0 and scored >= candidate_target:
            break
    return best.results()
```

There is one important distinction here.

Stopping simply because we have collected, for example, 100 candidates is an approximate stopping rule. It does not mathematically prove that a better document does not exist outside the current prefix.

Therefore, in the experiments, recall still needs to be measured against the same exact reference results. An exact version would require a valid upper bound on the score values of documents outside the currently visited prefix.

Relationship to HNSW

At first glance, this approach may sound somewhat similar to HNSW because both methods use a hierarchy to progressively change the search space.

However, I think that the hierarchy is being used differently.

[top -> bottom] HNSW builds a proximity graph. During a query, it begins from sparse upper layers, navigates toward promising neighbors, and progressively descends toward the denser bottom layer [2].

[bottom -> top] Backward Walk does not need to navigate between neighboring document vectors. The query itself directly tells us where to begin:

query $\rightarrow$ binary prefix $\rightarrow$ exact prefix lookup

We then broaden that prefix only when necessary ("from the bottom, going to the top").

Potential advantage 1: cheaper index construction and updates

One possible advantage that I think might happen (and would like to investigate further) is the cost of adding new documents.

For HNSW, inserting a new vector generally requires searching the existing graph to find suitable neighbors and then creating or updating graph connections.

For Backward Walk, the conceptual insertion process is much simpler:

document $\rightarrow$ embedding $\rightarrow$ binary signs $\rightarrow$ prefix key $\rightarrow$ corresponding bucket

There is no nearest-neighbor graph that needs to be constructed for each new document. So Backward Walk may be attractive for a search index that changes frequently.

I would frame this as a hypothesis to verify, because the actual insertion cost still depends on how the buckets, ranges and on-disk/in-memory index are implemented.

Potential advantage 2: Matryoshka prefixes give the relaxation some meaning

There is also a useful connection with Matryoshka embeddings.

A Matryoshka model is trained so that shorter prefixes of the embedding remain useful representations [3].

Therefore, moving from, for example, 16 bits $\rightarrow$ 15 bits $\rightarrow$ 14 bits is not simply dropping arbitrary dimensions from an ordinary embedding. We are progressively moving to shorter prefixes of a representation that was specifically trained to retain useful information at multiple prefix lengths.

However, this does not mean that the first coordinate is always more important than the second, or that the nearest neighbor under 16 dimensions must remain the nearest neighbor under 15 dimensions.

Matryoshka training gives us a reason to believe that shorter prefixes remain meaningful; the actual recall of each prefix still has to be measured.

2 Preliminary calculations

Before running the experiments, let's first break down what's the time/memory complexity for each method.

to recap:

- Method A performs a document scoring operation for every document inside the opened clusters.
- Method B avoids some document scoring operations by filtering with Bitplanes first.
- Method C avoids starting with a large document set by looking up a specific prefix, then broadening only when needed.

So for each method, I mainly want to understand two things:

- What additional work do Methods B and C introduce?
- How many document scoring operations can Methods B and C potentially save?

Let's use this as variables:

Symbol	Meaning
N	Total number of documents
C	Total number of clusters
P	Number of clusters opened for one query (P is for "probes")
d	Number of binary dimensions
K	Number of results we want to return
n_c	Number of documents inside cluster c
F_A, F_B, F_C	Number of documents that undergo a document scoring operation under A, B and C

If the clusters are roughly balanced, one cluster contains about

$$n \approx \frac{N}{C}$$

documents.

So if we open P clusters, we expose roughly

$$\frac{PN}{C}$$

documents.

2.1 Query-time work

Let's try to imagine this using example. Suppose one opened cluster contains three documents:

ID	Binary document	Full score value
0	10100	0.1
1	11001	2.5
2	11000	1.9

The query is

$$q = [0.7, 0.5, -0.4, -0.6, 0.3].$$

From the previous section, we know that the ideal binary document is

11001

and suppose we want the best

$$K = 2$$

documents.

For this example:

$$N = 3, \quad C = P = 1, \quad d = 5.$$

*note: the actual experiments later contain hundreds of thousands or millions of documents.

2.1.1 Method A: scan the opened clusters

Method A is the easiest one to understand.

Once cluster selection (prefilter) has decided which clusters to open, **for every document inside those clusters, we will do a scoring operation.**

Step 1: how many document scoring operations do we need?

In our example, the only opened cluster contains:

[ID 0, ID 1, ID 2]

so:

$$F_A = 3.$$

More generally, let O be the set of clusters opened for one query. Then:

$$F_A = \sum_{c \in O} n_c$$

is the number of documents for which A performs scoring operations.

If the clusters are roughly balanced:

$$F_A \approx \frac{PN}{C}.$$

So for A, this is already the most important number: **how many documents did cluster selection leave for us to search?**

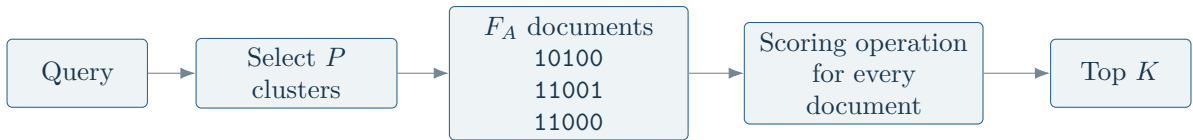


Figure 8: Method A performs no additional filtering after cluster selection. Every document inside the opened clusters undergoes a document scoring operation.

Step 2: what does one document scoring operation cost?

The scorer used here follows the four-sign lookup-table approach described in Exa's article.

Instead of multiplying every query value with every document bit individually, the query is divided into groups (or subvectors) of four.

For our five-dimensional example:

$$\begin{bmatrix} 0.7, & 0.5, & -0.4, & -0.6 \\ 0.3, & 0, & 0, & 0 \end{bmatrix}$$

A four-bit group has

$$2^4 = 16$$

possible binary patterns.

So this example needs

$$2 \times 16 = 32$$

precomputed table entries.

In general, the number of groups is

$$G = \left\lceil \frac{d}{4} \right\rceil.$$

One document scoring operation therefore reads G table entries.

For our example:

$$G = 2.$$

Three documents therefore require:

$$3 \times 2 = 6$$

table lookups.

In general, the scoring work grows with

$$F_A G.$$

Since G grows with d ,

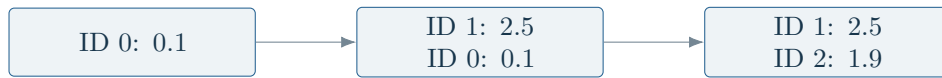
$$T_{\text{score},A} = O(F_A d).$$

For a fixed binary dimension, this is basically linear in the number of documents for which we perform scoring operations.

Step 3: keep only the best K

As score values are produced, we only need to keep the best K .

For our example:



Using a bounded heap, maintaining the best K results takes at most approximately

$$O(F_A \log(K + 1)).$$

Total cost for A

Let T_{base} contain the work before document scoring operations, such as cluster selection and lookup-table preparation.

Let a_A be the average measured time needed to fully process one document, including its scoring operation and the top- K update.

Then:

$$T_A \approx T_{\text{base}} + F_A a_A.$$

For roughly balanced clusters:

$$T_A \approx T_{\text{base}} + \frac{PN}{C} a_A.$$

The main tradeoff is straightforward.

Using more clusters usually makes each cluster smaller, so each opened cluster contains fewer documents. However, we may then need to open more clusters to maintain the same recall.

TL;DR — Method A: A's cost is mainly determined by how many documents cluster selection leaves behind. Fewer opened documents means fewer document scoring operations, but opening too few clusters can hurt recall.

2.1.2 Method B: Bitplanes

Bitplanes try to reduce the number of documents that undergo a document scoring operation.

Instead of immediately performing a scoring operation for every document inside the opened clusters, B first uses Bitplanes to filter the documents.

So B has two main costs:

$$\text{Bitplane work} + \text{scoring operations for the documents that survive}$$

The whole idea only makes sense if the document scoring operations we avoid are more expensive than the Bitplane work required to avoid them.

Step 1: inspect the important query bits first

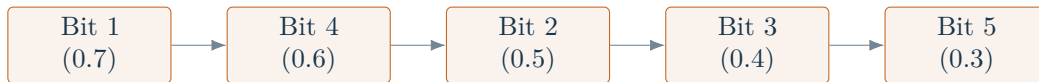
B uses the same cluster selection and the same lookup table as A.

The additional step is to order the query bits by decreasing $|q_i|$.

For

$$q = [0.7, 0.5, -0.4, -0.6, 0.3],$$

the order is:



We inspect the larger values first because mismatching them hurts the score value more.

Sorting these positions costs

$$O(d \log d).$$

We also calculate the ideal score value:

$$Q = \sum_i |q_i| = 2.5.$$

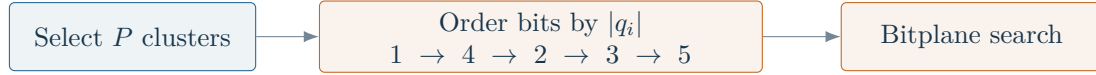


Figure 9: Bitplanes keep the same cluster-selection stage as A, but add query-dependent filtering before the full scorer.

Step 2: represent the documents using a mask

Inside one cluster, B keeps a bitmap showing which document positions are still active.

If cluster c contains n_c documents, its mask needs

$$W_c = \left\lceil \frac{n_c}{64} \right\rceil$$

64-bit machine words.

Our three-document example fits inside one word.

The starting mask is:

111

which means that all three documents are still active.

Step 3: split using the query bits

The first query position is bit 1, and the query prefers 1.

All three documents match:

$$111 \text{ AND } 111 = 111$$

so nothing is removed.

The next position is bit 4. The query prefers 0, and again all three documents match.

So the mask is still:

111.

The next position is bit 2.

Its Bitplane is:

011.

Therefore:

$$111 \text{ AND } 011 = 011$$

The preferred branch now contains IDs 1 and 2.

The alternative branch is:

100

which contains ID 0.

Because ID 0 disagrees with the query at bit 2, this branch has mismatch penalty:

$$p = |q_2| = 0.5.$$

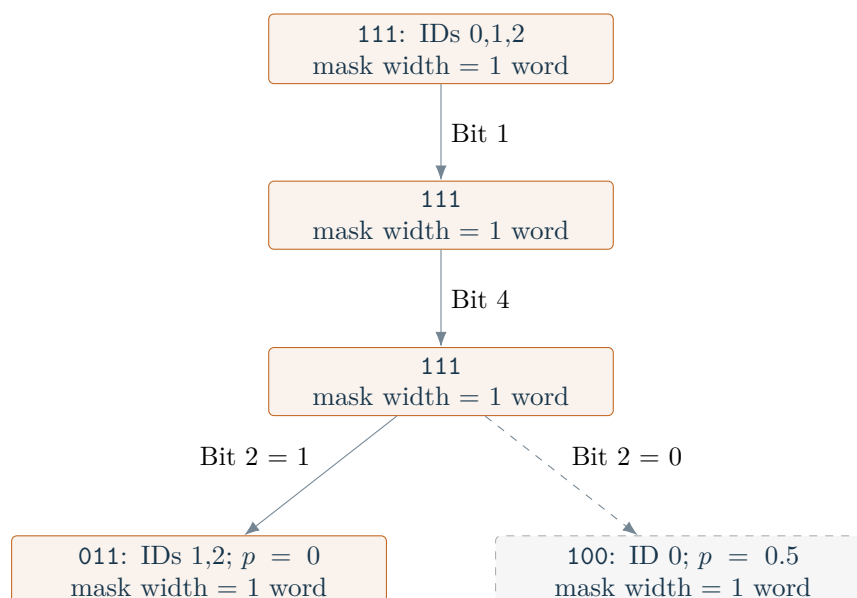


Figure 10: The first two bit tests remove no documents. The third separates ID 0 into another branch, but that branch is not discarded yet.

Step 4: why Bitplane filtering is not $O(1)$ - it's not free! :(

At first, a bitwise AND seems like it's $O(1)$.

But actually it's not. Example -> If one cluster contains:

6,400

documents.

Then, for every one mask we need:

$$6400/64 = 100$$

machine words.

So one split does not perform one AND, as we need to loop across roughly 100 word positions.

There is another important detail: **the mask does not physically get shorter when fewer documents remain active.**

For example, suppose only 80 documents survive after several splits. But in this case, the mask can still be 100 words wide.

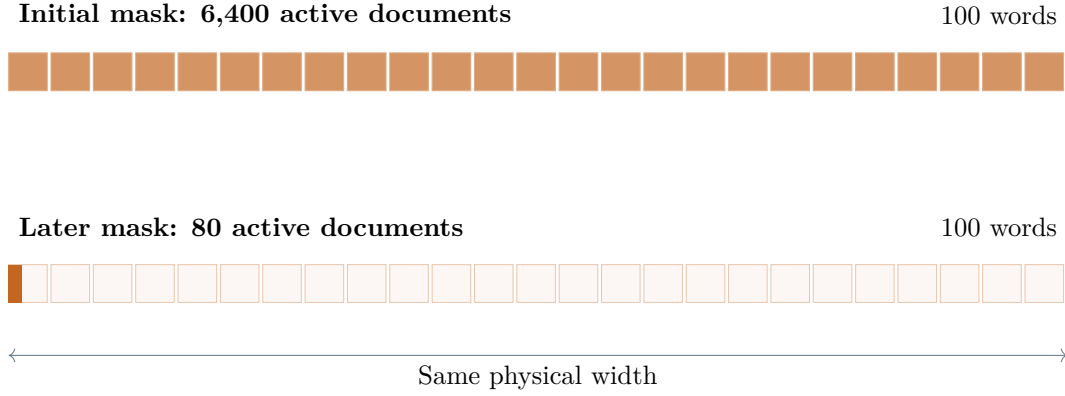


Figure 11: The number of active documents can become much smaller while the physical bitmap stays the same size.

Step 5: count the Bitplane work

Let J_c be the number of times B splits a mask in cluster c .

Each split scans W_c words.

So the total amount of split-mask work is:

$$V_{\text{split}} = \sum_{c \in O} J_c W_c.$$

For roughly balanced clusters, if B performs an average of S splits per opened cluster:

$$V_{\text{split}} \approx PS \left\lceil \frac{N}{64C} \right\rceil.$$

Step 6: eventually stop splitting and perform scoring operations

After a branch contains few enough documents, we treat this as a **leaf**, and in this case, method B will perform document scoring operations for the documents inside it.

Suppose our leaf size is two.

The preferred branch:

011

contains exactly IDs 1 and 2.

So B performs these document scoring operations:

$$\begin{aligned} \text{ID 1} &\rightarrow 2.5 \\ \text{ID 2} &\rightarrow 1.9 \end{aligned}$$

Therefore:

$$F_B = 2.$$

*in this case, if we do a comparison, method A would have performed three document scoring operations.

another little cost: we still need to read leaf masks

Before B can extract the surviving document positions, it still has to scan the leaf mask. Let H_c be the number of leaf masks read in cluster c .

$$V_{\text{leaf}} = \sum_{c \in O} H_c W_c.$$

Step 7: the un-explored branch that is still in the queue

Remember when we consider 'branching' to catch all cases because we are worried that we might miss the ideal case due to branching? Imagine that: ID 0 is still waiting to be explored.

We can calculate the known mismatch penalty, which is:

$$p = 0.5.$$

However, we can take a look that the best possible score value inside that branch is:

$$Q - 2p = 2.5 - 2(0.5) = 1.5.$$

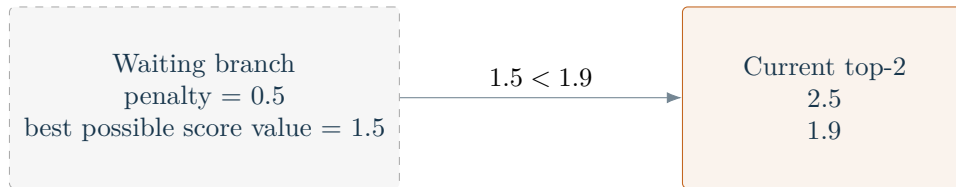
Meanwhile, our current top two score values are already:

$$2.5, \quad 1.9.$$

Since:

$$1.5 < 1.9,$$

We can make conclusion: nothing inside the waiting branch can enter the current top two. So the good news is that, we can safely skip it!



→ skip branch

Figure 12: Once the best possible score value of a branch falls below the current top- K threshold, the whole branch can be skipped.

Total cost for B

B therefore pays for:

- ordering the query bits;
- reading split masks;
- reading leaf masks;
- maintaining the waiting branches;
- performing F_B document scoring operations.

Let:

Symbol	Meaning
b_s	Average time for one split-mask word
b_l	Average time for one leaf-mask word
T_{order}	Ordering query bits and preparing query weights
T_{queue}	Managing branches, masks and the priority queue
a_B	Average time to fully process one surviving document

Then:

$$T_B \approx T_{\text{base}} + T_{\text{order}} + b_s V_{\text{split}} + b_l V_{\text{leaf}} + F_B a_B + T_{\text{queue}}.$$

The key condition is:

$$\text{document scoring work avoided} > \text{Bitplane work added.}$$

A smaller leaf size may reduce F_B , because B keeps filtering longer. However, that also means more splits and more bitmap work.

A larger node budget can also improve recall, but again it allows B to search more branches.

TL;DR — Bitplanes: B is useful when a relatively small amount of bitmap work can eliminate a large number of document scoring operations. Performing fewer document scoring operations alone is not enough; the bitmap work also has to be cheap enough.

2.1.3 Method C: Backward Walk

Now let's look at the second idea.

Backward Walk approaches the problem from almost the opposite direction.

Method A starts with every document inside the opened clusters.

Method B also starts with all of them, but progressively removes documents.

Backward Walk instead asks:

What if we start from the most specific binary prefix first, and only broaden the search when we need more documents?

So C mainly pays for two things:

- looking up different prefixes;
- performing scoring operations for the documents exposed by those prefixes.

Step 1: store the documents in prefix order

Using the same example, let's sort the three documents by their binary key:

Position	Binary key	ID
0	10100	0
1	11000	2
2	11001	1

Meanwhile, the ideal query pattern is 11001.

Because the rows are sorted, documents with the same prefix will appear next to each other.

That means we can use binary search to quickly find the range corresponding to a particular prefix.

Step 2: start from the most specific prefix

Let's first use all five bits: 11001.

Two binary searches find the beginning and the end of this range.

In our example, the result is $[2, 3)$, which contains only ID 1.

So we perform one scoring operation: $\text{ID 1} \rightarrow 2.5$.

However, remember that we want the top two results. Right now we only have one document.

So unfortunately, we cannot stop yet. We need to broaden the prefix.

Step 3: remove one constraint

We now go from 11001 to 1100*.

The * means that we no longer care whether the last bit is 0 or 1.

The new range becomes $[1, 3)$, which contains:

- ID 2: 11000;
- ID 1: 11001.

Now, ID 1 was already scored at the previous depth.

So the good news is that we do **not** need to score it again.

The only new document is ID 2: $\text{ID 2} \rightarrow 1.9$.

Now we have two results.

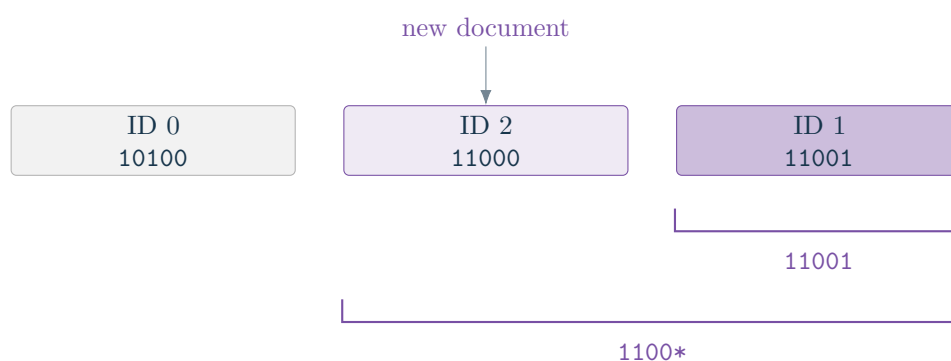


Figure 13: The broader prefix contains the previous range. ID 1 is already scored, so only the newly included document, ID 2, needs a scoring operation.

Step 4: how much prefix lookup work do we need?

Suppose we start at prefix depth x , and eventually stop at depth ℓ .

Then the number of depths we visit is:

$$L = x - \ell + 1.$$

At every positive depth, we need two binary searches for every opened cluster:

- one to find where the prefix starts;

- one to find where it ends.

So if we open P clusters and visit L_+ positive prefix depths, we perform $2PL_+$ binary searches.

Depth zero uses the known whole-cluster range and needs no binary search.

Each binary search over cluster c costs roughly:

$$O(\log(n_c + 1)).$$

So the total lookup work is approximately:

$$2L_+ \sum_{c \in O} \log_2(n_c + 1).$$

Again, the formula looks more complicated than the idea.

The idea is simply:

Backward Walk cost grows with how many prefix depths we visit and how many clusters we search at each depth.

Step 5: how many documents do we eventually score?

Let F_C be the total number of distinct documents that become visible before we stop.

Because each broader prefix contains the previous prefix, a document that was already scored does not need another scoring operation.

So C's document-scoring work is simply:

$$F_C a_C.$$

Here a_C is the average time for a document scoring operation and its top- K update.

This means the main question for Backward Walk is actually:

How broad does the prefix need to become before we have enough useful documents?

If we can stop while the prefix is still very specific, F_C can be very small.

If we need to keep broadening all the way to the root, F_C eventually becomes all documents in the opened clusters.

Step 6: how quickly does the prefix grow?

Let's use a simple model just to build intuition.

Assume, temporarily, that every bit is independent and has a 50% chance of being 0 or 1.

Then the probability that a random document matches an ℓ -bit prefix is:

$$2^{-\ell}.$$

So if we have n documents, the expected number matching that prefix is:

$$\frac{n}{2^\ell}.$$

For a collection of one million documents:

Prefix depth	Expected documents
16	about 15
15	about 31
14	about 61
13	about 122
12	about 244

So something interesting happens here.

We only remove **one bit** each time, but the number of matching documents approximately **doubles**.

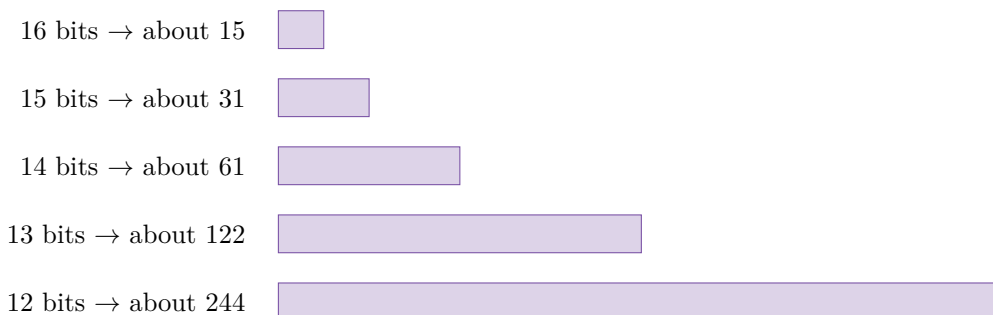


Figure 14: Under the independent 50/50-bit model, removing one prefix bit doubles the expected matching count. The labels are rounded; they describe document counts, not recall.

However, we should remember that this is only an intuition.

Real embedding bits can be correlated, and once the documents have already been divided into clusters, the distribution can be even further from a perfect 50/50 split.

Most importantly:

This only estimates how many documents we might find. It does not tell us whether those documents are actually the correct top results.

Step 7: when should Backward Walk stop?

There are two ways to stop.

Approximate stopping

The simplest option is to stop after we have found enough documents.

For example:

16 bits	→ 12 documents
15 bits	→ 31 documents
14 bits	→ 66 documents
13 bits	→ 136 documents

If our target is 100 documents, we could stop at 13 bits.

This is simple and cheap. However, there is one problem.

Finding 100 documents does not prove that the best unseen document is worse than the ones we found.

So this stopping rule is approximate, and we need to measure its recall experimentally.

Exact stopping

Fortunately, we can also use the scoring equation to derive a safe bound.

Suppose we have already searched all documents matching the first ℓ query signs inside the opened clusters.

Any unseen document in those clusters must disagree with the query on at least one of those first ℓ positions.

The cheapest possible mismatch is:

$$m_\ell = \min_{1 \leq i \leq \ell} |q_i|.$$

Therefore, the best possible score of an unseen document is:

$$U_\ell = Q - 2m_\ell.$$

Meanwhile, suppose we have already kept K documents and the worst document currently inside our top- K has score τ .

If:

$$\tau > U_\ell,$$

then even the **best possible unseen document** cannot enter our top- K .

So the good news is that we can stop safely. The implementation also keeps its tie-breaking and floating-point rounding checks.

If this never happens, we can keep broadening the prefix until depth zero.

At depth zero, we have exposed every document inside the opened clusters, so Backward Walk becomes equivalent to Method A locally.

Total cost for Backward Walk

Backward Walk therefore pays for:

- building the query prefix;
- performing prefix lookups;
- updating the current prefix ranges;
- performing scoring operations for F_C exposed documents.

We can summarize this as:

$$T_C \approx T_{\text{base}} + T_{\text{key}} + 2L_+ t_b \sum_{c \in O} \log_2(n_c + 1) + F_C a_C + T_{\text{ranges}}.$$

T_{key} covers the query key and optional bound preparation; t_b is time per binary-search comparison; T_{ranges} covers range updates and stopping checks.

Again, the exact symbols are not the most important part.

The important condition is:

$\text{document scoring work avoided} > \text{prefix lookup work added}$

If C can stop at a fairly deep prefix, then $F_C \ll F_A$, and we may save a lot of scoring work.

However, if C has to broaden the prefix all the way to depth zero, $F_C = F_A$, then C ends up doing all of A's scoring operations **plus** the additional prefix lookups.

TL;DR — Backward Walk: Backward Walk works best when a fairly specific prefix already contains the documents we need. If we have to keep broadening until almost the whole cluster becomes visible, then the prefix lookup has not saved us anything.

2.2 Memory complexity

So far, we have only talked about query time.

However, B and C also need additional index structures, so we should also check how much memory they require.

There are two kinds of memory that matter here:

- **persistent index memory**, which remains loaded between queries;
- **temporary query memory**, which only exists while one query is running.

2.2.1 Method A memory

Method A stores:

- the packed binary document;
- the document ID;
- the cluster/routing information.

Using 64-bit words, one binary document needs

$$8 \left\lceil \frac{d}{64} \right\rceil$$

bytes. The ID adds another 8 bytes.

So the document storage is:

$$M_{A,\text{rows}} = N \left(8 \left\lceil \frac{d}{64} \right\rceil + 8 \right).$$

During a query, A also keeps the query values, the lookup table and the current top- K .

This is our baseline.

2.2.2 Method B memory

B keeps everything A needs.

However, it adds two things. First, it stores the Bitplanes.

For each cluster, every binary dimension needs one bitmap spanning all document positions in that cluster.

Therefore:

$$M_{\text{planes}} = 8d \sum_{c=1}^C W_c.$$

So in a rough sense, B stores the document bits twice:

- once arranged by document;
- once arranged as Bitplanes.

The second extra cost is temporary query memory.

Remember that branching means we can have several masks waiting in the queue at the same time.

Each one still has the full physical width of its cluster.

If $A_c(t)$ masks belonging to cluster c are alive at time t , then the raw mask memory is:

$$Q_{\text{masks}} = 8 \max_t \sum_c A_c(t) W_c.$$

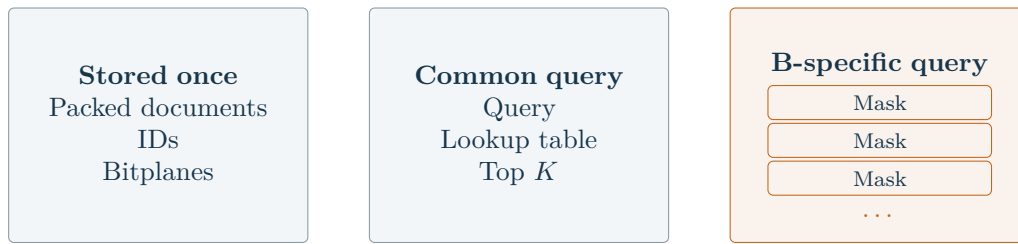


Figure 15: B stores Bitplanes alongside its packed documents. During a query, several full-width masks may also remain alive in the branch queue.

TL;DR — B memory: Bitplanes need extra persistent memory because we store another arrangement of the document bits. During search, B can also temporarily hold several full-width branch masks at the same time.

2.2.3 Method C memory

C also keeps the normal packed documents and IDs.

The current implementation additionally stores one 32-bit prefix key for every document. So:

$$M_{\text{prefix}} = 4N.$$

The important part is that we do **not** create a new copy of the documents for every prefix depth. For example, 11001, 1100*, 110** all refer to different ranges of the **same sorted document array**.

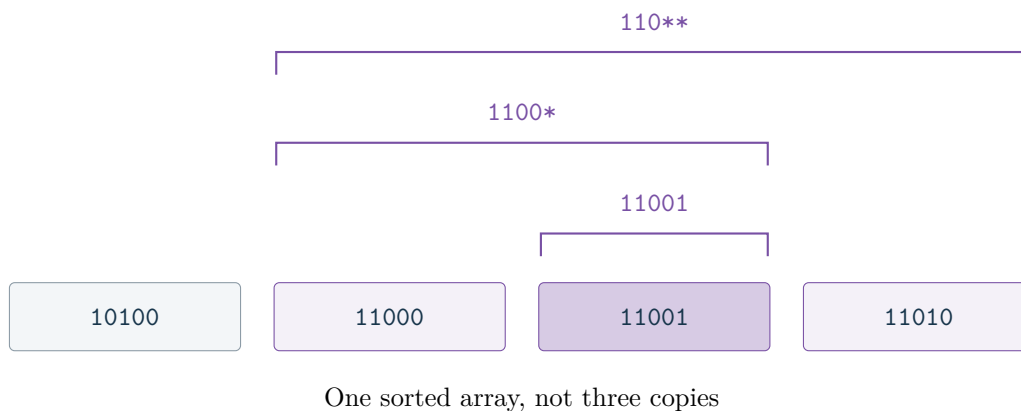


Figure 16: In this illustrative array, three prefix ranges refer to the same stored rows. Broadening the prefix does not create a new document list.

During the query, C mainly needs to remember the previous range for every opened cluster.

So compared with B, its temporary search state is relatively small. Actual byte totals still depend on the selected parameters and number of live branches.

TL;DR — C memory: Backward Walk adds a small prefix key per document, but it does not need full-width masks for many waiting branches.

2.3 Recall

Now we know roughly how much work each method performs.

However, none of those savings matter if we stop too early and return the wrong documents. So we also need to understand where recall can be lost.

For query q :

- G_q is the exact global top- K ;
- $O_{a,q}$ is the result returned by method a .

Average recall is:

$$R_a = \frac{\sum_q |G_q \cap O_{a,q}|}{n_q K}.$$

Here n_q is the number of queries. The reference and returned results use the same score and document-ID tie rule.

For example:

Reference: [1, 2]
Returned: [1, 0]

means we recovered one of two: 50%.

Where can the correct document disappear?

There are actually two places.

First, cluster selection can fail to open the cluster containing the correct document.

Second, even if the correct document is inside an opened cluster, B or C may stop their local search before reaching it.

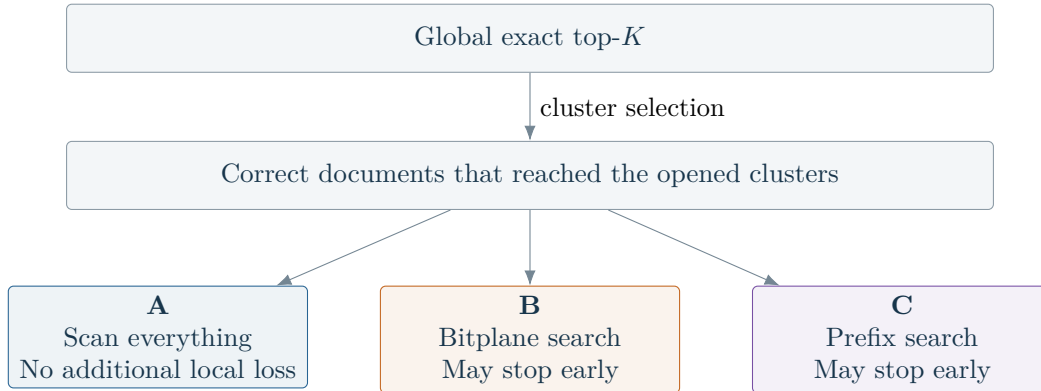


Figure 17: Cluster selection can lose a correct document before local search begins. B and C may also lose a document by stopping their local search early.

2.3.1 Recall for Method A

Let g_q be the number of global top- K documents whose clusters were opened.

Because A scores every document inside those clusters, all g_q of those documents are recovered locally. So:

$$R_{A,q} = \frac{g_q}{K}.$$

In other words:

A can lose a correct document because cluster selection missed it, but once the document is inside an opened cluster, A will not lose it during local search.

2.3.2 Recall for Bitplanes

Suppose g_q correct documents reached the opened clusters, but B only recovers some fraction $r_{B,q}$ of them.

Then:

$$R_{B,q} = \frac{g_q}{K} r_{B,q}.$$

If $g_q = 0$, recall is zero.

For example, imagine:

- 8 of the global top 10 documents are inside the opened clusters;
- B recovers 6 of those 8.

Then:

$$\text{Recall} = \frac{8}{10} \times \frac{6}{8} = 60\%.$$

If B performs a complete branch-and-bound search using valid score bounds, then it eventually recovers all of the correct documents inside the opened clusters.

However, if we impose a finite node budget, B may stop early.

TL;DR — B recall: B can lose recall in two places: cluster selection can miss the document before B starts, and a limited Bitplane search can miss a document even after its cluster has been opened.

2.3.3 Recall for Backward Walk

For Backward Walk, let $D_{q,\ell}$ be all documents exposed when the search stops at prefix depth ℓ .

Then:

$$R_{C,q} = \frac{|G_q \cap D_{q,\ell}|}{K}.$$

The important thing here is that **candidate count and recall are not the same thing**.

Finding 100 documents does not mean that we found the correct 100 documents.

Consider:

$$q = [0.1, 0.9].$$

The ideal binary pattern is 11 and suppose the only documents are 10 and 01.

Their scores are:

$$S(10) = -0.8, \quad S(01) = 0.8.$$

If we search prefix 1*, we find 10.

However, the better document is 01, which is outside that prefix.

So even though we found one candidate, we did not find the best result.

That is why stopping after “enough documents” is only an approximate rule.

C is exact locally only when:

1. we broaden all the way to depth zero and expose every document inside the opened clusters;
or
2. the score bound proves that none of the unseen documents can beat the current top- K .

TL;DR — C recall: A large prefix range does not automatically mean high recall. Backward Walk can safely stop only when it has either searched every document or mathematically proved that the unseen documents cannot win.

2.4 Summary

The whole calculation section can be summarized with one simple idea:

Method	What it does
A	Score every document inside the opened clusters
B	Spend Bitplane work to avoid some document scoring
C	Spend prefix-lookup work to avoid some document scoring

So:

A just pays for document scoring. B and C deliberately add extra search work, hoping that this lets them avoid enough document scoring to make the total query faster.

The experiment in the next section will tell us whether that actually happens on real embeddings.

3 Experiments

So far, we have only looked at the three methods mathematically.

The calculations tell us what *could* make Bitplanes or Backward Walk faster, but the actual result depends on the documents, the queries, and how much work each method ends up doing.

So the experiment is simple:

For the same search problem and the same recall requirement, what is the fastest configuration I can find for A, B and C?

Each method is allowed to choose the parameters that make sense for it.

Method A can choose its cluster count and how many clusters to open.

Method B can choose those same cluster settings, together with its leaf size and node budget.

Method C can choose the cluster settings, starting prefix depth and stopping rule.

The documents, queries, binary scoring function and correct answers remain unchanged.

This is important because B and C do not necessarily work best with the same cluster layout as A. For example, Bitplanes may be able to open a much larger group of documents and then remove most of them before the document scoring operation.

3.1 Setup

I use two collections:

Collection	Embedding model	Documents	Queries	Binary dimensions
MS MARCO	Nomic v1.5	1,000,000	1,000	64, 256, 768
Quora	Qwen3 0.6B	522,931	1,000	32, 256, 1024

The document and query embeddings are generated before the search experiment begins.

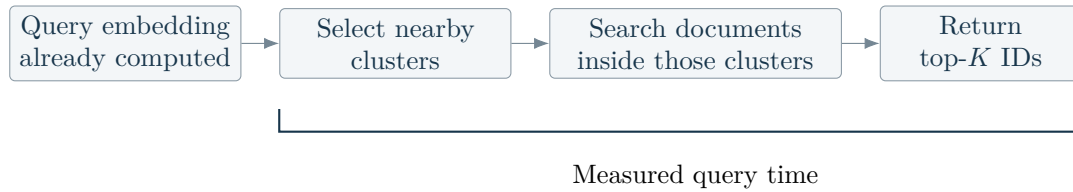


Figure 18: The query embedding is already available when timing begins. Query time includes cluster selection, search inside those clusters and returned IDs.

The timer therefore measures only the part of the pipeline that I am trying to improve: Embedding generation and the later full-vector reranking stage are not included.

For one comparison, A, B and C use:

- the same documents;
- the same 1,000 queries;
- the same binary dimension;
- the same scoring function;
- the same requested result count K ;
- the same exact reference answers.

The idea is simply: **same search problem, different way of searching it.**

Changing the binary dimension is treated as a different comparison because, for example, a 32-bit binary vector and a 256-bit binary vector do not necessarily rank the documents in the same order.

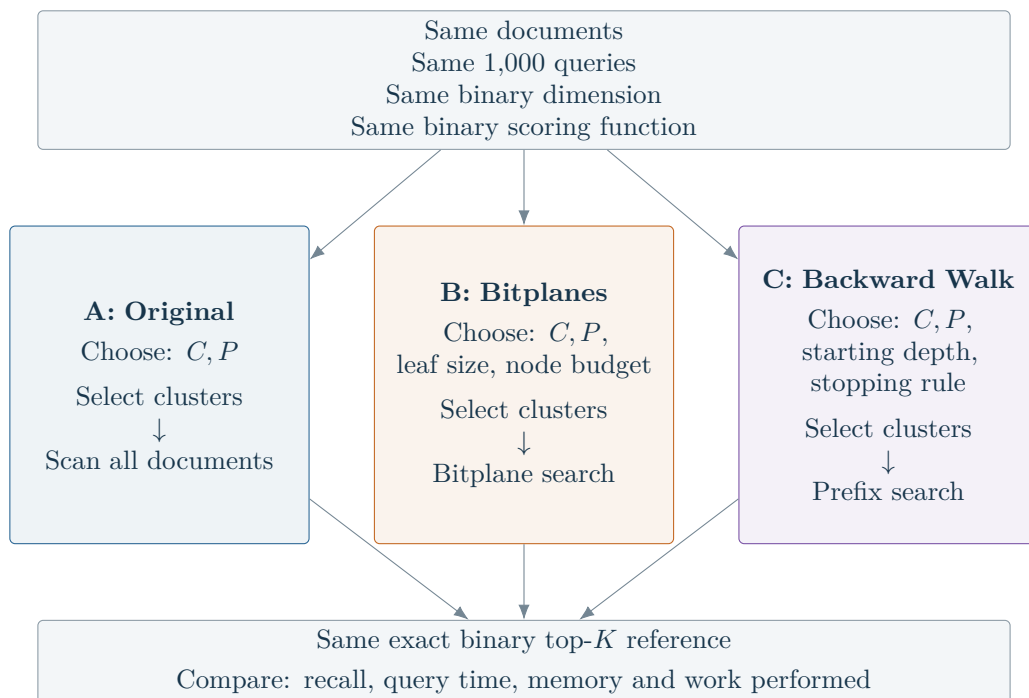


Figure 19: A, B and C search the same documents and queries using the same binary scoring function. Each method can choose its own index and search parameters; the fastest configuration meeting the same recall requirement is compared.

3.2 How the comparison works

Before testing approximate search, I first calculate the exact answer for every query.

I do this by performing a scoring operation for every document in the collection using the same binary scoring function used by A, B and C.

Suppose the exact top five document IDs are [12, 7, 91, 4, 30] and one method returns [12, 7, 91, 4, 85].

Four of the five correct IDs were recovered, so:

$$\text{Recall@5} = 80\%.$$

I repeat this over all 1,000 queries.

Therefore, in this document, if it's written as **99% recall**, we mean that the method recovers 99% of the exact binary top- K document IDs on average.

Choosing the parameters

For every method, we try multiple configurations.

For A:

- number of clusters C ;
- number of opened clusters P .

For B:

- C ;
- P ;
- leaf size;
- node budget.

For C:

- C ;
- P ;
- starting prefix depth;
- stopping rule.

For a given recall target, such as 80%, 90%, 95%, 99%, I discard configurations that do not reach that recall.

Among the remaining configurations, I keep the fastest one for each method.

So the comparison is:

3.3 Main result

Let's first look at Qwen with 32-bit binary documents while returning one result: $K = 1$.

This is the setting where the difference between the three search strategies is the clearest.

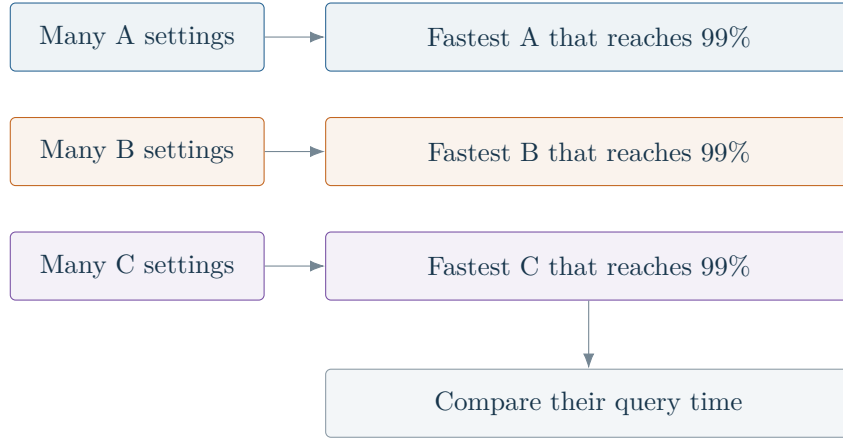


Figure 20: At each recall target, compare the fastest tested configuration for each method that meets the requirement. Measurements below the target remain in the saved results, but cannot win that comparison.

Required recall	A: Original	B: Bitplanes	C: Backward Walk
80%	0.0308 ms	0.0295 ms	0.0318 ms
90%	0.0498 ms	0.0432 ms	0.0497 ms
95%	0.0805 ms	0.0520 ms	0.0799 ms
99%	0.1948 ms	0.0809 ms	0.1688 ms

At the 99% recall target, Bitplanes take 0.0809 ms compared with 0.1948 ms for A.

So in this setting, B is approximately 2.4× faster than the original method.

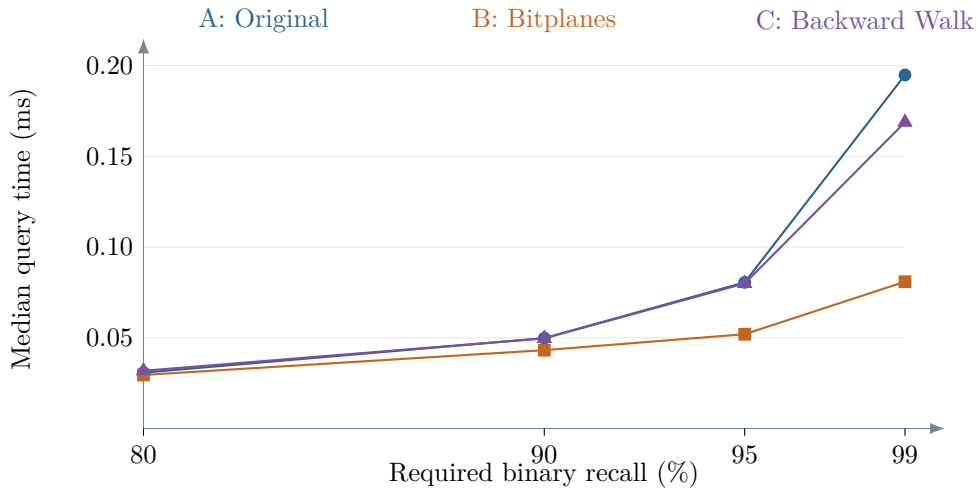


Figure 21: Query time for Qwen32 with $K = 1$. Each point shows the repeated median for the fastest tested configuration selected at the required binary recall.

3.4 Why is Bitplanes faster?

let's look at the recall 99% settings chosen by each method.

Method	Clusters	Clusters opened	Search setting	Recall	Time (ms)
A	4096	180	Scan	99.0%	0.1948
B	16	9	Leaf 128, budget 8192	99.2%	0.0809
C	1024	78	Start depth 4, exact stop	99.0%	0.1688

The first interesting thing is that the methods choose very different cluster layouts.

Now we can also observe how many documents reach the search stage and how many of them undergo a document scoring operation:

Method	Documents inside opened clusters	Scoring operations
A	24,181	24,181
B	295,031	818
C	41,147	26,153

We can observe the following:

Method A

A uses very selective cluster selection:

select clusters $\rightarrow$ 24,181 documents $\rightarrow$ 24,181 scoring operations.

Method B

B does almost the opposite:

select much larger clusters $\rightarrow$ 295,031 documents
 $\downarrow$ Bitplanes
 818 document scoring operations.

B therefore performs scoring operations for only

$$818/295031 \approx 0.28\%$$

of the documents inside its opened clusters.

This is exactly the tradeoff described in Section 2. B performs extra bitmap work, but in this case that work removes so many document scoring operations that the complete query becomes faster.

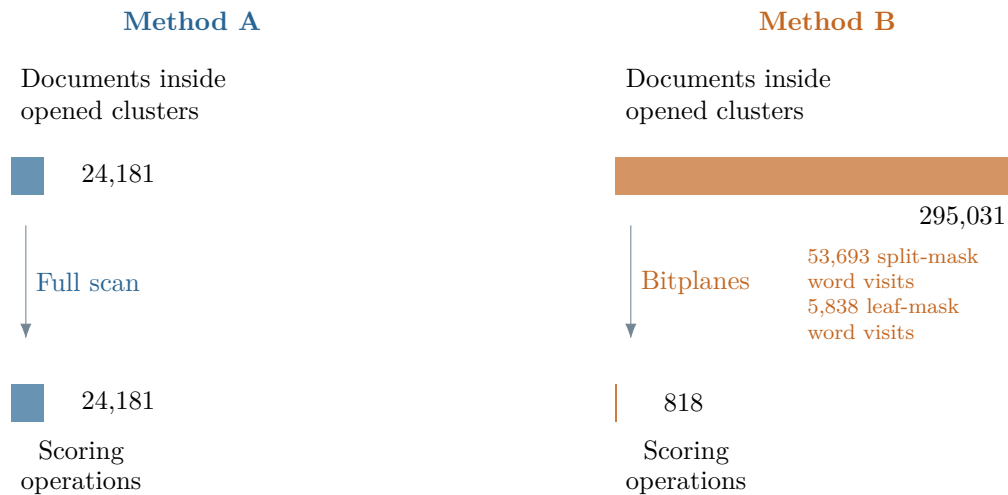


Figure 22: At the selected 99% setting, A opens a small set of documents and performs scoring operations for all of them. B opens a much larger set but uses Bitplanes to reduce roughly 295,000 documents to only 818 document scoring operations.

What about C?

Backward Walk also removes some document scoring operations:



Figure 23: C reduces the number of document scoring operations, but exposes a larger fraction of its selected documents than B does.

However, the reduction is much smaller than B’s. C also needs to perform prefix lookups while progressively broadening the query prefix.

At this setting, C visits around 3.5 prefix depths and performs around 495 boundary searches per query.

Its final time is 0.1688 ms, which is better than A in this particular comparison, but still much slower than B.

So the three methods are effectively making three different choices:

A	Reduce documents heavily during cluster selection, then perform a scoring operation for every document
B	Use coarse cluster selection, then aggressively prune with Bitplanes
C	Use medium cluster selection, then progressively broaden a prefix

3.5 Does this advantage always happen?

TLDR: Not really. The previous result asks for only one final result: $K = 1$. Now let’s say, we increase the number of requested results to $K = 100$.

At the 99% binary recall target:

Representation	A	B	C
Nomic 64	1.915 ms	2.071 ms	1.875 ms
Nomic 256	13.453 ms	13.866 ms	14.645 ms
Nomic 768	45.088 ms	45.217 ms	45.196 ms
Qwen 32	0.400 ms	0.439 ms	0.404 ms
Qwen 256	3.564 ms	3.721 ms	4.039 ms
Qwen 1024	22.863 ms	23.098 ms	22.930 ms

In these selected settings, B performs no useful splitting and C starts from depth zero. In other words, both methods effectively fall back toward the original scan.

C’s slightly lower median on Nomic64 is a scan-like timing difference; it does not come from avoiding document scoring operations.

This also makes intuitive sense. When we only need one result, B can search the most promising branches and safely ignore many alternatives.

When we need 100 results, many more branches can still contain one of the required documents. B therefore has to search much more of the tree.

Eventually, the cost of pruning is no longer worth it.

The Bitplane advantage therefore appears strongest when K is small enough that many branches can be safely ignored.

This is a measured result for the tested settings, not a guarantee that B can never help for larger K .

3.6 Summary

For Qwen32 with one requested result, Bitplanes are clearly faster in the tested search setup.

At 99% binary recall:

- A takes 0.1948 ms;
- B takes 0.0809 ms;
- C takes 0.1688 ms.

B is therefore about $2.4\times$ faster than A.

Why?.

A selects a relatively small number of documents and then performs scoring operations for all of them.

B selects a much larger number of documents, but its Bitplane search reduces roughly 295,000 opened documents to only around 818 document scoring operations.

In this setting, the document scoring work avoided is large enough to repay the extra bitmap work.

However, this does not happen everywhere. When the result count increases to $K = 100$, B and C become much closer to ordinary scanning and the advantage disappears in the tested settings.

So the result is not: **Bitplanes are always faster.**

It is: **Bitplanes can be faster when the query only needs a small number of results and the query bits allow a large fraction of the opened documents to be ruled out before document scoring operations.**

Backward Walk shows the same general tradeoff, but in the tested data it does not reduce the scoring work as aggressively as Bitplanes.

4 Limitations and next steps

The timings describe this CPU implementation. All three methods reuse the same C++ lookup-table scorer, which adds entries using float64. Exa's production system and its SIMD kernels are not reproduced here. A faster scoring kernel can change the balance between document scoring operations and maintaining an index search.

I hold the collection, model and code length fixed when choosing each method's parameters. Shorter codes change the reference ranking, so their binary recall does not establish the same retrieval quality as a longer representation. The source datasets provide human relevance labels for that separate evaluation.

These are static indexes built before querying. The experiment does not measure document updates or concurrent requests. In particular, keeping rows sorted still costs work when new documents are added. The memory formulas count the binary search index and its query buffers; an application that keeps uncompressed vectors or document text in RAM must add that storage.

Prefix expansion also appears in LSH Forest [8] and PUFFINN [9]. Those methods use multiple hash indexes and different stopping rules. Their recall guarantees do not transfer to a single prefix in the embedding's original coordinate order.

5 Conclusion

For Qwen32 with one returned result, B takes 0.081 ms at 99.2% binary recall, versus A’s 0.195 ms and C’s 0.169 ms at 99.0%. All methods can choose binary-trained clusters. With 100 results at the 99% target, the selected B and C settings instead become scans. C has a narrower exact-search benefit: it takes 4.875 μ s on the 82 queries whose complete code exists, while B remains faster over all 1,000 queries.

These results identify useful tested settings, not a universal optimum. Qwen32’s exact binary top 1 agrees with the full-float best score value on only 14% of queries, so the short-code speed does not preserve full-embedding quality. For a fixed representation and recall requirement, the decision remains whether the scoring operations avoided pay for the added mask or prefix work. The measured counts let us check that explanation directly.

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